

GENAI Internship - Project 3 Milestone

Team Project: Web Knowledge Bot with PDF RAG

DEADLINE: 29-06-2026 , 07:00pm

1. Project Overview

Welcome to the **third and final project milestone** of the GENAI Internship. This is a **team-based project** where you will collaborate to build a **Web Knowledge Bot**—an AI-powered chatbot that answers questions based on the content of a specific website, with the added capability of uploading and querying PDF documents.

2. Core Concept: Web Knowledge Bot + PDF RAG

What is a Web Knowledge Bot?

It's a chatbot that lives on a website and answers questions about that website's content. Think of it as a customer support or FAQ bot that knows everything about the site it's embedded on.

Example: If you build a bot for a college website, the bot should answer questions like:

- "What are the admission requirements?"
- "How can I contact the faculty?"
- "What are the course offerings?"

The RAG PDF Addition:

Users should be able to upload a PDF document and ask questions about it, alongside the website content. The bot should combine knowledge from both sources.

3. Team Assignments & Project Focus

Team Name	Members	Website Domain Focus	Specific Task
Gryffindor	Raviteja, Tharuni, Harika	E-commerce Website	Products, pricing, shipping, returns
Slytherin	Priyadarshini, Vikranth, Sushma	Legal Services Site	Services, resources, contact
Ravenclaw	Ramya, Shiva Patel, Vaishnavi Sony	NNRG College Website	Campus information, admissions, academics
Hufflepuff	Sindhuja, Vyshnavi, Shiva Kumar	Tech Documentation Site	API docs, developer guides, SDKs
Alpha	Surya, Varshitha, Arya	Healthcare/Medical Site	Services, appointments, health tips
Beta	Jathin, Maruthi, Praveen	Travel/Hospitality Site	Destinations, bookings, packages
Gamma	Anusha, Vijitesh, Bharghav	Food/Restaurant Website	Menu, reservations, delivery

4. The Two Components

Component 1: Website Knowledge Bot

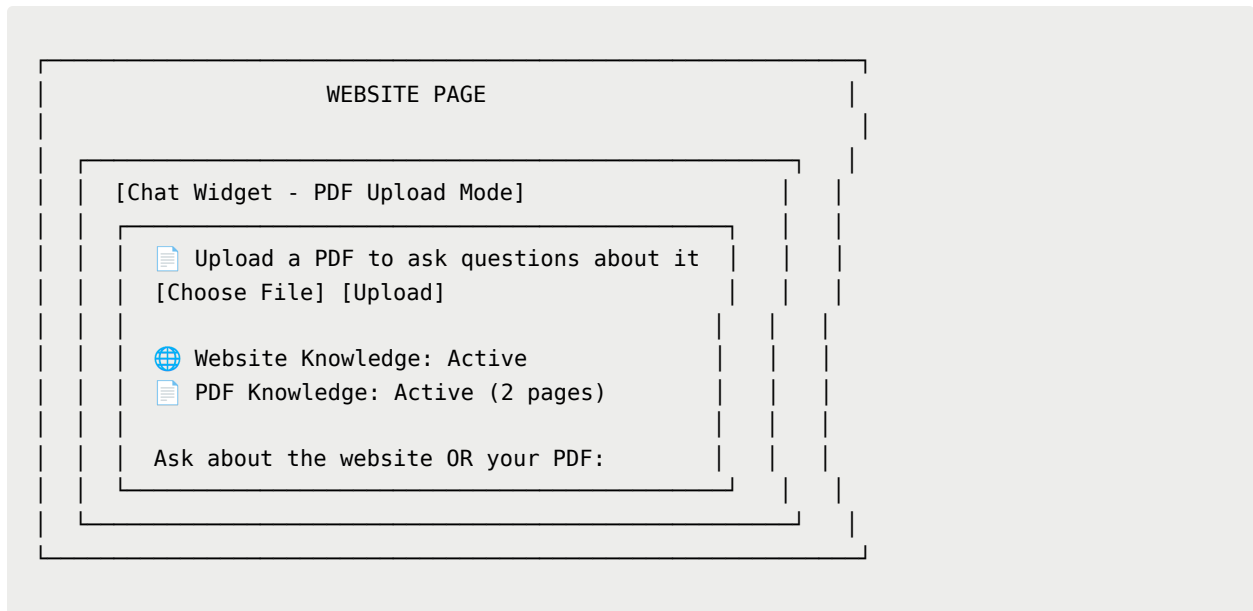
The chatbot is embedded on a website (as a widget or floating button) and answers questions **exclusively about that website's content**.

text



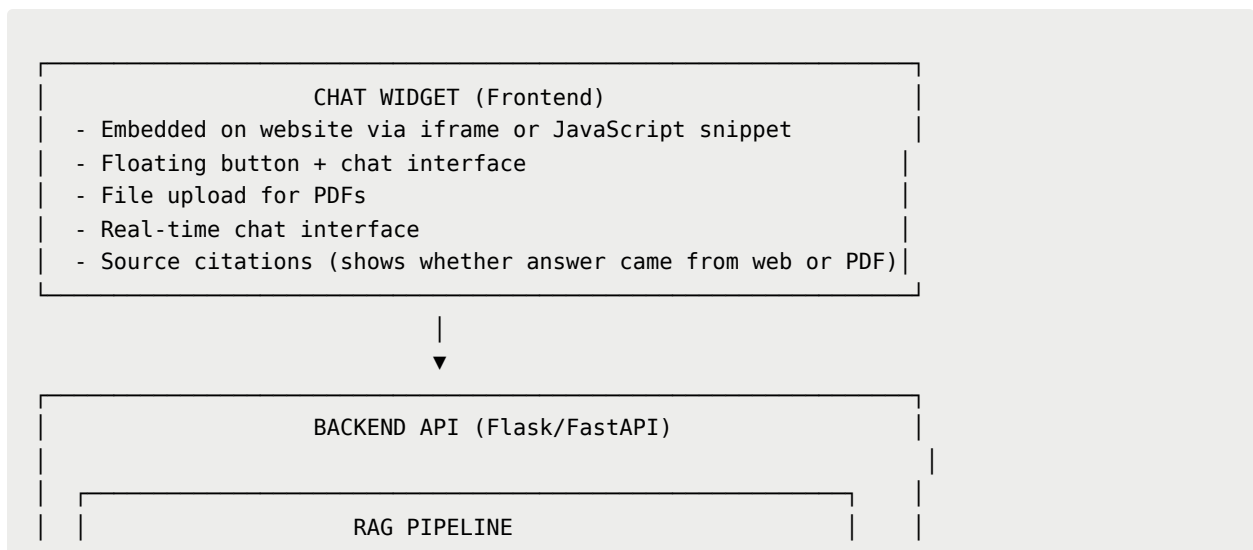
Component 2: PDF RAG Integration

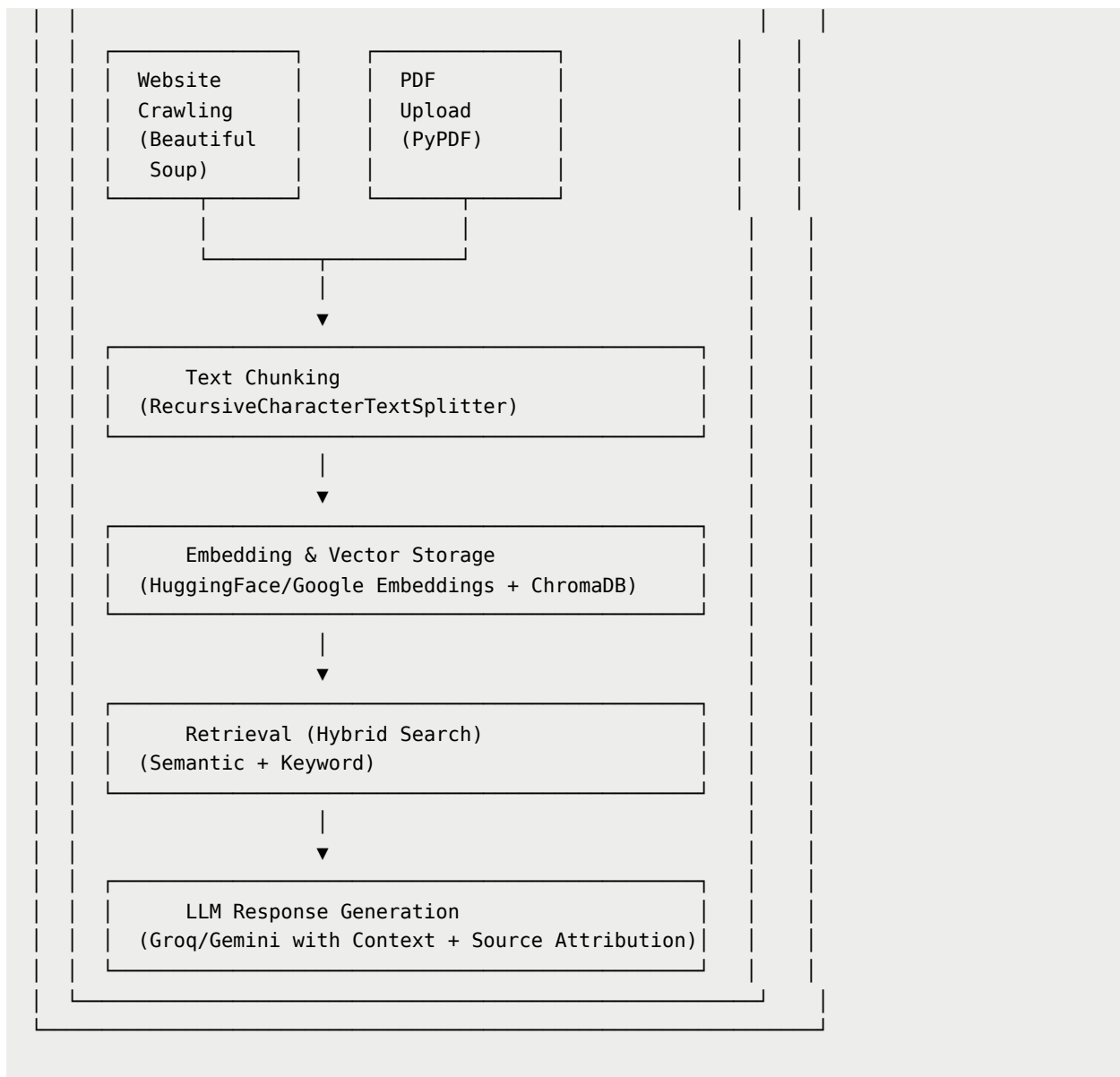
Within the same widget, users can upload a PDF and ask questions about it.
text



5. Technical Architecture

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6. Required Features

6.1 Core Features

Feature	Description	Technical Implementation
Website Chat Widget	Floating chatbot on the website	JavaScript widget + CSS
Website Knowledge Base	Index the target website content	BeautifulSoup + LangChain WebBaseLoader

Feature	Description	Technical Implementation
PDF Upload & Processing	Upload and index PDF content	PyPDF + TextSplitter
Unified RAG Pipeline	Answer from both sources	LangChain + Vector DB
Source Attribution	Show whether answer came from web or PDF	Metadata tracking
Example Questions	Pre-filled questions for users	UI component

6.2 Bonus Features

Feature	Description
Multi-LLM Support	Groq, Gemini, or OpenAI selection
Hybrid Search	Semantic + Keyword search
Out-of-Domain Handling	Polite decline for unrelated questions
Chat History	Session-based conversation memory
Download Chat	Export conversation as CSV

7. Technical Requirements

7.1 Tech Stack Options

Component	Technology	Reason
Frontend	Streamlit or React + Widget	For chat UI and embeddable widget
Backend	FastAPI or Flask	API for chat and RAG pipeline
LLM	Groq or Gemini	Fast, generous free tiers
Vector DB	ChromaDB or FAISS	Local, fast, efficient
PDF Parsing	PyPDF	Lightweight, reliable
Web Scraping	BeautifulSoup + LangChain WebBaseLoader	Standard, effective

7.2 Project Structure

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```
web-knowledge-bot/
├── frontend/
│   ├── widget.js          # Embeddable chat widget
│   ├── widget.css        # Widget styling
│   └── index.html         # Demo page
├── backend/
│   ├── app.py             # Main API
│   ├── crawler.py        # Website crawling
│   ├── pdf_processor.py   # PDF processing
│   ├── rag_pipeline.py    # RAG logic
│   ├── vector_store.py    # ChromaDB operations
│   └── config.py          # API keys and settings
├── README.md              # Documentation
└── requirements.txt        # Dependencies
```

8. User Experience Flow

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1. User visits website
- ↓
2. Sees chat widget in bottom-right corner
- ↓
3. Clicks "Ask a question" or types a message
- ↓
4. User can:
 - a. Ask a question about the website content
 - b. Upload a PDF and ask questions about it
- ↓
5. Bot answers with:
 - a. Response text
 - b. Source attribution (Web / PDF)
 - c. Confidence or citation
- ↓
6. Bot also suggests follow-up questions

9. Why This Project Matters

This project combines the two most important GenAI application areas for enterprises:

Aspect	Why It's Valuable
Website Knowledge Bot	Automates customer support, reduces response time
PDF RAG Integration	Handles complex documents (reports, manuals, legal docs)
Combined Capability	One bot for all company knowledge—website + documents

Real-World Impact:

- Reduced support tickets
- 24/7 availability
- Consistent answers across all channels
- Scalable knowledge management

10. Submission Requirements

10.1 What to Submit

Item	Deadline
Public URL (live deployed bot)	29-06-2026, 7:00 PM IST
GitHub Repository Link	29-06-2026, 7:00 PM IST
Team Name & Project Title	29-06-2026, 7:00 PM IST
Team Members Names	29-06-2026, 7:00 PM IST

10.2 README Requirements

Your README must include:

Section	Content
Project Title & Team	Team name, project name, members
Overview	What your Web Knowledge Bot does
Architecture Diagram	Visual representation of your system
Setup Instructions	Steps to run locally
API Keys Required	Groq/Gemini setup
Deployment	How to deploy the bot

Section	Content
Team Contributions	Who worked on what

11. Point System & Scoring

11.1 Base Points (Maximum 100 Points)

Component	Criteria	Points
A. Submission & Deployment (15 pts)		
A1	URL submitted before deadline	5
A2	Chat widget is accessible	5
A3	Bot responds to test prompts	3
A4	Deployment is stable	2
B. Frontend Requirements (10 pts)		
B1	Chat widget embedded on website	5
B2	PDF upload + website knowledge source display	3
B3	Footer with team name & milestone info	2
C. Website Knowledge (20 pts)		
C1	Website content indexed successfully	8
C2	Bot answers questions from website	7
C3	Out-of-domain questions handled	5
D. PDF RAG Integration (20 pts)		
D1	PDF upload and processing works	8
D2	Bot answers questions from PDF	7
D3	Source attribution shows PDF vs Web	5
E. Live Demo (15 pts)		
E1	Demonstrates 3+ working examples	10

Component	Criteria	Points
E2	Clear explanation of RAG pipeline	5
F. Q&A Session (10 pts)		
F1	Answers technical questions about RAG	10
G. Documentation (10 pts)		
G1	README is complete with all sections	5
G2	Architecture diagram and setup instructions	5
Total Base Points		100

11.2 Bonus Points (Top Teams)

Rank	Bonus Points
1st Place	+50 bonus points
2nd Place	+25 bonus points
3rd Place	+15 bonus points

12. Code of Conduct

Rule	Requirement
Team Collaboration	All team members must contribute
Individual Work	No copying from other teams
Domain Adherence	Must follow assigned website domain
Deadline Respect	Late submission = point deduction
Cameras On	During demos and closing ceremony

13. Final Message

"This is your final milestone—a team project where you'll build a Web Knowledge Bot with PDF RAG capabilities. Combine everything you've learned: chatbots, RAG systems, document processing, and frontend embedding. Build

| *something that could actually be used in the real world. We're here to support you. Good luck!"*

- **Sreekar, Karthik, Divya Anu, Aditya, Balaji**